

Name: _____ Date: _____

Period: _____

Warm-Up — Key

1. Solve $x^2 + y^2 = 100$ for y . How many solutions for y are there when $x = 6$?

$$y = \pm\sqrt{100 - x^2}$$

Number of solutions: **2**

2. The table below shows (x, y) pairs on a curve.

x	y
1	3
2	5
4	9

Compute $\Delta y/\Delta x$ from $x = 1$ to $x = 2$: $(5 - 3)/(2 - 1) = 2$

Is this ratio positive or negative? **Positive**

What does the sign tell you about how y changes as x increases? **As x increases, y also increases (same direction).**

3. The equation is $y^2 = x + 4$.

(a) For $x = 0$, what are the y -values? $y = \pm 2$

(b) For $x = 5$, what are the y -values? $y = \pm 3$

Teacher Note

Q1: Students must write $\pm$; a common error is writing only the positive root. Emphasize that both are valid solutions.

Q2: The ratio $\Delta y/\Delta x = 2$ is positive, meaning x and y increase together. Use this to preview co-variation vocabulary in the lesson.

Q3: Remind students to use the square-root property. For $x = 5$: $y^2 = 9$, so $y = \pm 3$.